

# GPS versus Local Knowledge Student Handout

TOK Cognitive Science Activity Bank • Knowledge and Technology • 25 min

|            |                          |      |                            |
|------------|--------------------------|------|----------------------------|
| Category   | Knowledge and Technology | Time | 25 min                     |
| Difficulty | Easy                     | ID   | gps-versus-local-knowledge |

Big idea: Data can be precise and incomplete.

## Student-Facing Prompt

Inspect GPS versus Local Knowledge Stimulus A. Record one first claim, the evidence you used, your confidence, and what would make you revise.

## Concrete Stimulus Packet

|                                                  |                                                                                                                                                                                                                                 |                                                                                                    |
|--------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| <b>GPS versus Local Knowledge Stimulus A</b>     | Student-facing stimulus 1: A route-planning scenario where the map suggests a faster but impractical route. Rank, sort, or audit this output. Identify which rule, ranking signal, or interface choice shaped your judgement.   | Students inspect this exact card first and commit to a claim before hearing anyone else.           |
| <b>GPS versus Local Knowledge Stimulus B</b>     | Student-facing stimulus 2: Local knowledge cards: school gate closed, roadworks, unsafe crossing, steep hill. Rank, sort, or audit this output. Identify which rule, ranking signal, or interface choice shaped your judgement. | Use this for comparison, a second round, or a partner challenge.                                   |
| <b>GPS versus Local Knowledge Reveal / Twist</b> | Student-facing stimulus 3: A decision table balancing speed, safety, reliability, and lived experience. Rank, sort, or audit this output. Identify which rule, ranking signal, or interface choice shaped your judgement.       | Teacher reveal: connect the changed response to the big idea — Data can be precise and incomplete. |

## Actual Lesson Questions

| # | Question for students                                                                       |
|---|---------------------------------------------------------------------------------------------|
| 1 | After inspecting GPS versus Local Knowledge Stimulus A, what is your first knowledge claim? |
| 2 | Which exact detail from GPS versus Local Knowledge Stimulus A did you use as evidence?      |
| 3 | What did you infer beyond the evidence?                                                     |
| 4 | How confident are you: 50%, 60%, 70%, 80%, 90%, or 100%?                                    |
| 5 | What missing information would most improve the knowledge claim?                            |
| 6 | When is local knowledge better than technological knowledge?                                |

## Ready-to-Use Examples

- A route-planning scenario where the map suggests a faster but impractical route.
- Local knowledge cards: school gate closed, roadworks, unsafe crossing, steep hill.
- A decision table balancing speed, safety, reliability, and lived experience.

## Student Task

Use the stimulus cards to complete the observation/inference/evidence/confidence grid, then answer one TOK knowledge question connected to Knowledge and Technology.

## Activity Steps

1. Give students a route recommended by an app.
2. Add local knowledge: roadworks, locked gate, dangerous crossing, staircase, informal shortcut, crowding at certain times.
3. Groups design the best route for different users.
4. Compare algorithmic optimisation with human judgement.

## Observation and Evidence Record

Use this grid to keep observation, inference, evidence, and confidence separate.

| What I noticed | What I inferred | Evidence | Confidence |
|----------------|-----------------|----------|------------|
|                |                 |          |            |
|                |                 |          |            |
|                |                 |          |            |

## TOK Debrief

- Knowledge question: When is local knowledge better than technological knowledge?
- Does data capture everything relevant?
- Can technology make us less attentive knowers?

## Knowledge Questions

- When is local knowledge better than technological knowledge?
- Does data capture everything relevant?
- Can technology make us less attentive knowers?

## Transfer Example

For example, run this activity with a route-planning scenario where the map suggests a faster but impractical route. Have students commit to an initial claim, then reveal the change, hidden rule, alternative context, or comparison that makes the knowledge problem visible.

## Exit Ticket

- One thing I first treated as evidence was...
- My confidence changed because...
- A better knowledge question I can now ask is...